The Complete AI Landscape

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① THE AI FAMILY TREE

Each inner ring is a specialisation of the one outside it — plus what sits parallel at each level

PARALLEL TO AI (non-ML approaches)

- Rule-based / Expert Systems
- Search & Optimisation Algorithms
- Knowledge Graphs & Ontologies
- Robotic Process Automation (RPA)

PARALLEL TO ML (other DL alternatives)

- Classical / Statistical ML
- Probabilistic / Bayesian Models
- Ensemble Methods (XGBoost etc.)

PARALLEL TO DEEP LEARNING

- Discriminative Models (CNNs)
- Deep Reinforcement Learning
- Representation Learning

PARALLEL TO GENERATIVE AI

- Standalone LLMs (no tools)
- RAG Systems (retrieval only)
- Fine-tuned / Domain Models
- Multimodal Models

PARALLEL TO AGENTS — other agentic / near-agentic patterns

- Copilot / Assistant (assists, doesn't act)
- RAG + Tools (retrieves + calls APIs)
- Multi-Agent Systems (agents orchestrating agents)

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② THE FOUR ML PARADIGMS

How machines learn — the fundamental approaches underneath every AI product

01 SUPERVISED
- Learns from labelled input/output pairs
- Classification & regression tasks
- Most common in enterprise AI

e.g. fraud detection, credit scoring

02 UNSUPERVISED
- No labels — finds hidden structure
- Clustering, segmentation, anomalies
- Dimensionality reduction

e.g. customer segments, outlier detection

03 REINFORCEMENT
- Agent learns via reward & penalty
- Trial and error over many episodes
- Optimises long-run strategy

e.g. AlphaGo, trading bots

04 SELF-SUPERVISED
- Creates its own labels from raw data
- Foundation of modern LLMs
- Predict next token / masked word

e.g. GPT, Claude, BERT

③ AI CAPABILITY TYPES — What the system does

The strategic lens: match the business need to the right AI capability

05 PERCEPTION
- Computer vision — ima
- Speech & audio recogn
- Document OCR & scan

e.g. ID verification, defect detec

06 PREDICTION
- Forecasting future value
- Risk scoring & credit m
- Demand & churn predic

e.g. loan default, sales forecast

07 GENERATION
- Text, code, images, auc
- Summarisation & draftir
- Translation & transform

e.g. report drafting, code assist

08 REASONING
- Multi-step logic & planni
- Question answering ove
- Decision support & expl

e.g. regulatory assessment, RA

09 RECOMMENDA
- Ranking & personalisati
- Collaborative filtering
- Next-best-action engine

e.g. product recs, policy matchi

10 OPTIMISATION
- Operations research + ML
- Scheduling & routing
- Portfolio & resource allocatio

e.g. logistics, capital allocation

④ HOW AI IS DEPLOYED — The product form it takes

The same model can be deployed in very different ways — the form determines adoption, risk, and value

EMBEDDED AI
Silent auto-decisions in background
Fraud alerts, email filters, autocomplete

ANALYTICS & BI
Insight surfaces in dashboards
Risk dashboards, KPI forecasts, reports

COPILOT / ASSIST
Helps human complete a task faster
Code assist, document drafting, Q&A

AUTONOMOUS AGENT
Acts independently using tools
Research agent, monitoring & alert bot

PLATFORM / API
AI as shared infrastructure layer
OpenAI API, Anthropic API, Azure AI

⑤ STRATEGIC DECISION GUIDE — From client problem to AI approach

Your consulting translation layer: what the client says → what AI branch to explore

What the client says	AI archetype	Technical approach
"We want to automate document review"	Document Intelligence	Supervised ML + LLM extraction
"We want to reduce fraud losses"	Prediction + Anomaly	Supervised / Unsupervised ML
"We want a chatbot for staff queries"	Knowledge Assistant	RAG + LLM (Generative AI)
"We want to personalise recommendations"	Recommendation Engine	Collaborative filtering / ML
"We want AI to monitor our operations"	Monitoring Agent	Agents + Anomaly Detection
"We want to optimise our supply chain"	Optimisation	Operations Research + ML

The inner path (AI → ML → DL → GenAI → Agents) is one branch of a much wider landscape